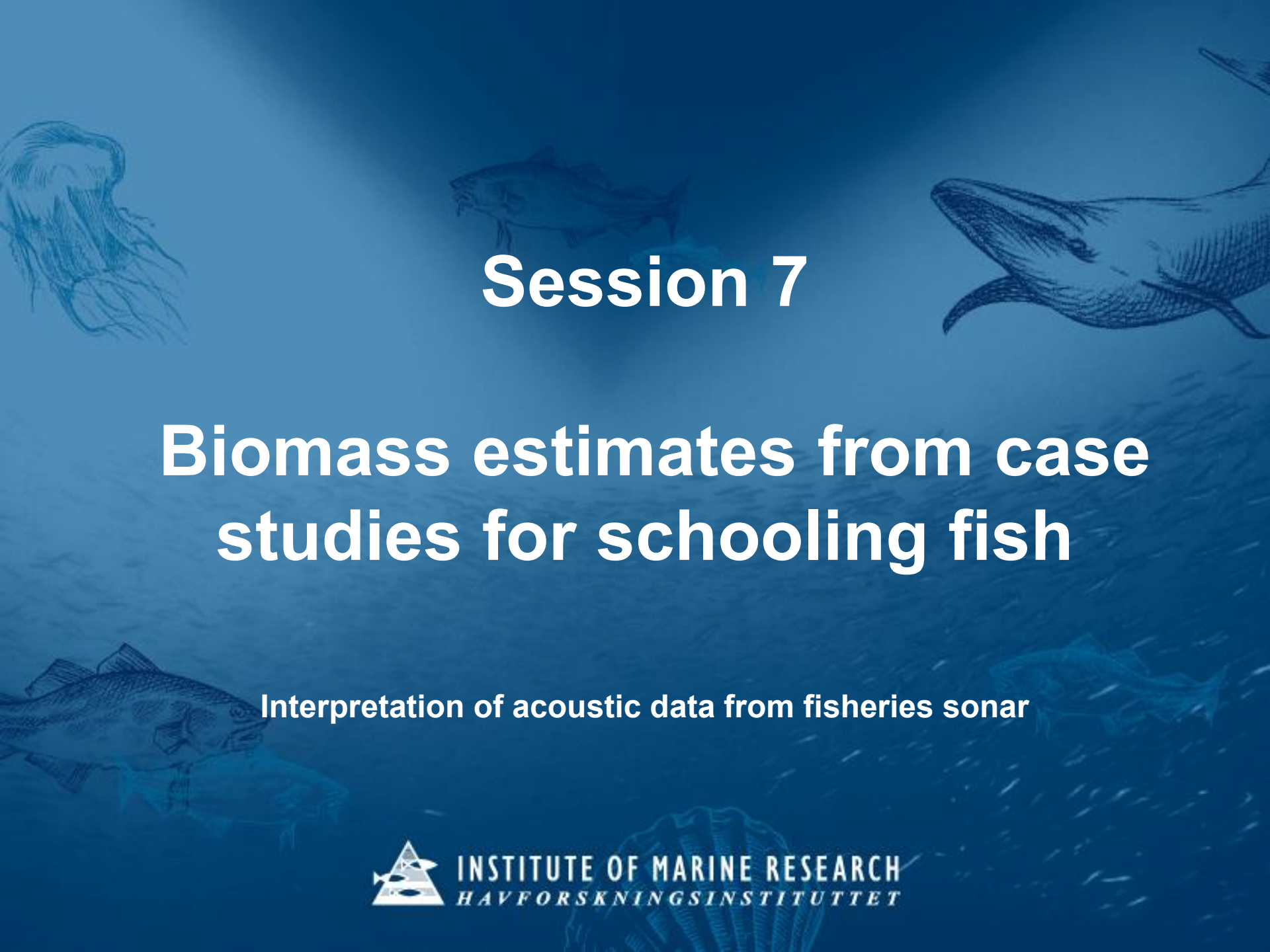




Session 7

Biomass estimates from case studies for schooling fish

Interpretation of acoustic data from fisheries sonar



INSTITUTE OF MARINE RESEARCH
HAVFORSKNINGSINSTITUTTET

Biomass estimates

1. Individual school estimates
 - Detailed inspection of single schools (< 5 knots)
2. Echo integration
 - During surveying (10-11 knots)



Individual school biomass

$$\text{school biomass} = \text{number of fish} * \text{individual fish weight}$$

Sonar data **Trawl data**

$$\text{number of fish} = \text{fish density (n/m}^3\text{)} * \text{school volume (m}^3\text{)}$$

$$\text{fish density} = \frac{\text{volume backscattering coefficient (s}_v\text{, m}^{-1}\text{)}}{\text{backscattering cross section (}\sigma_{bs}\text{, m}^2\text{)}}$$



Individual school biomass

$$\text{number of fish} = \text{school volume (m}^3\text{)} * \text{fish density (n/m}^3\text{)}$$

$$\text{fish density} = \frac{\text{volume backscattering coefficient (s}_v\text{, m}^{-1}\text{)}}{\text{backscattering cross section (}\sigma_{bs}\text{ , m}^2\text{)}}$$

$$\text{Fish_density} = \frac{s_v}{10^{(TS/10)}}$$



$$s_v = 10^{(S_v/10)}$$

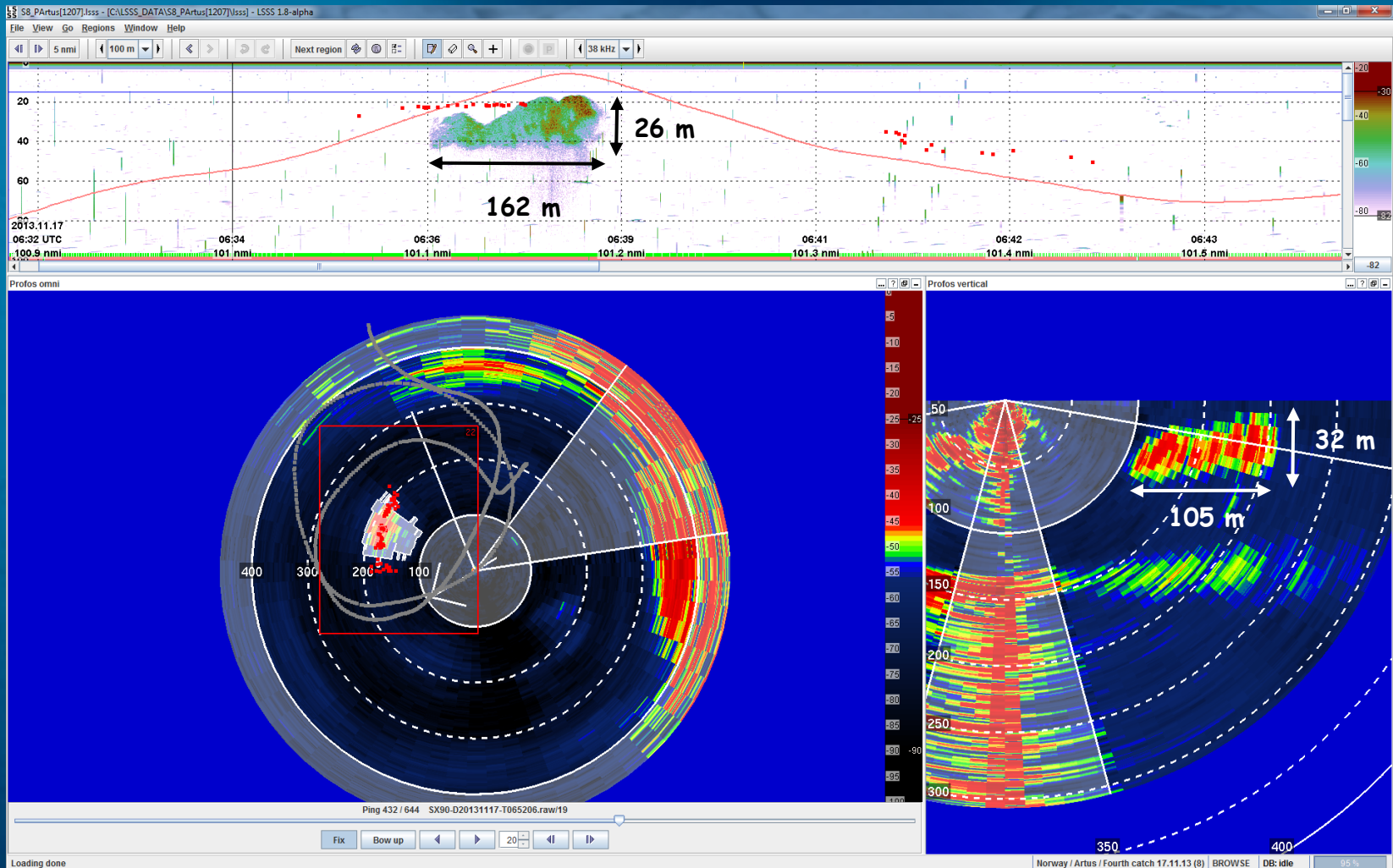
from sonar output we get:
 S_v , mean volume backscattering
strength (dB re 1 m⁻¹)



Side aspect
from *in situ* or models



Example using herring school in echo and vertical, November 2013



Example using herring school in echo and vertical, November 2013

Mean Sv from sonar

$$s_v = 10^{(S_v/10)}$$

$$s_v = 10^{(-38.1/10)}$$

$$\text{Fish_density} = \frac{s_v}{10^{(TS/10)}}$$

$$\text{Fish_density} = \frac{1.5E^{-4}}{8.0E^{-5}} = 1.9 \frac{\text{fish}}{\text{m}^3}$$

$$\sigma_{bs} = 10^{(TS/10)}$$

$$\sigma_{bs} = 10^{(-41.01/10)}$$

$$TS = 20 \log (35) - 71.9$$

$$\text{Number_of_fish} = (\text{school_volume})(\text{Fish_density})$$

$$\text{Number_of_fish} = 1.99E^{+5} 1.9$$

$$= 3.88E^{+5}$$

$$\text{School_biomass} = (\text{number_of_fish})(\text{mean_fish_weight})$$

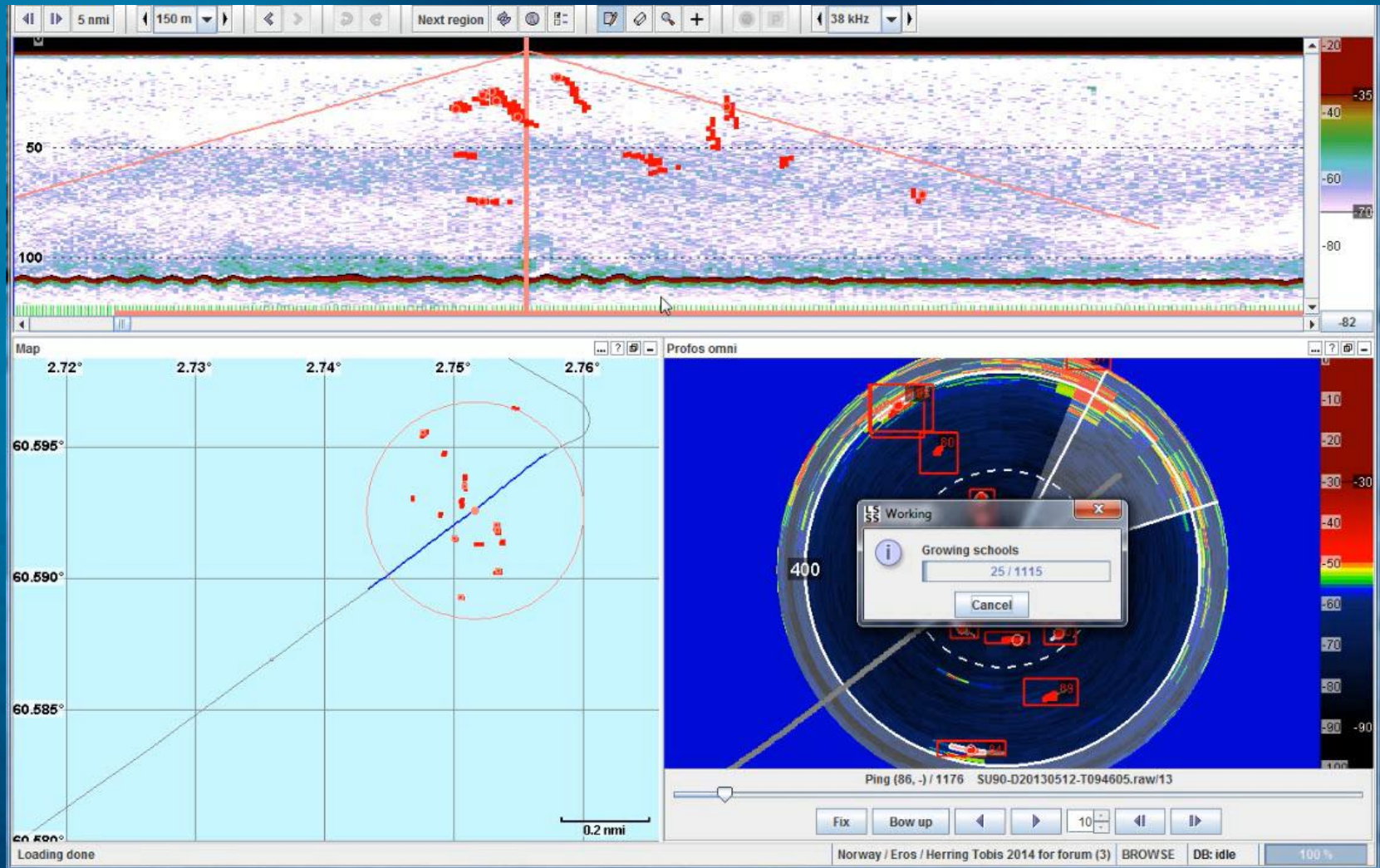
Mean fish weight = 340 g

$$\text{School_biomass} = 132 \text{ ton}$$

Real catch 110 ton

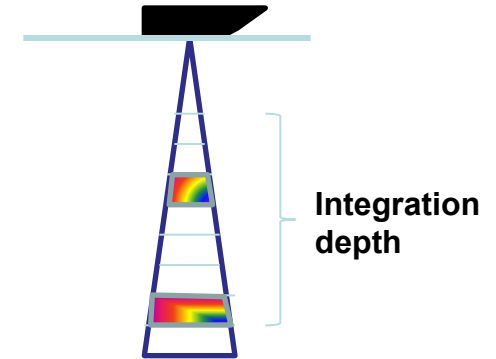
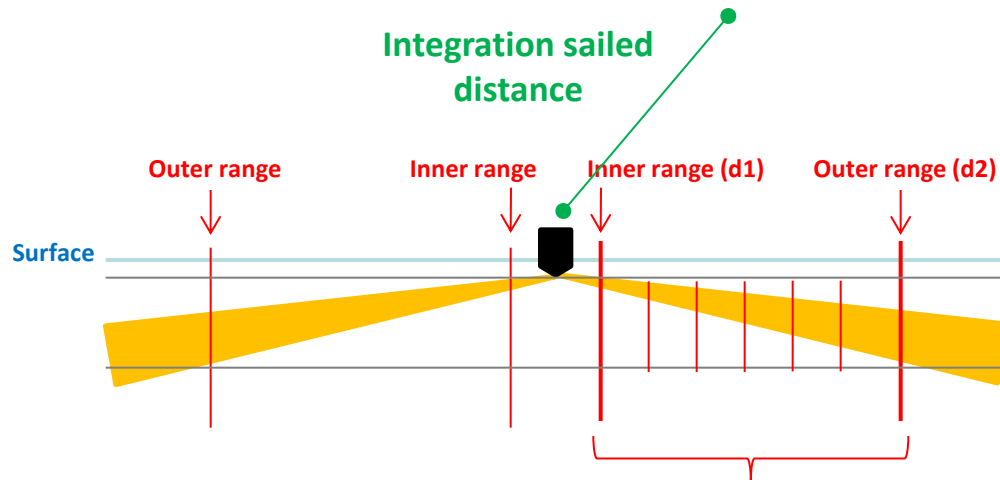


2. Echo integration of sonar data



2. Echo integration of sonar data

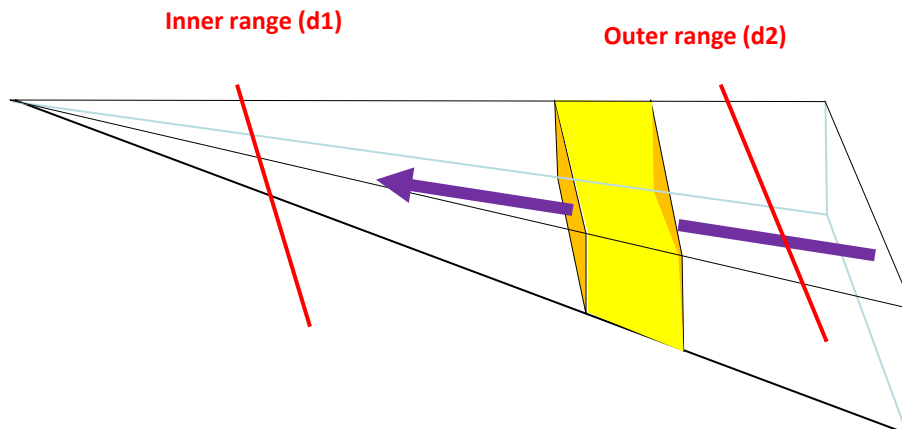
Echo sounder integration



Min. depth range (z1)

Max. depth range (z2)

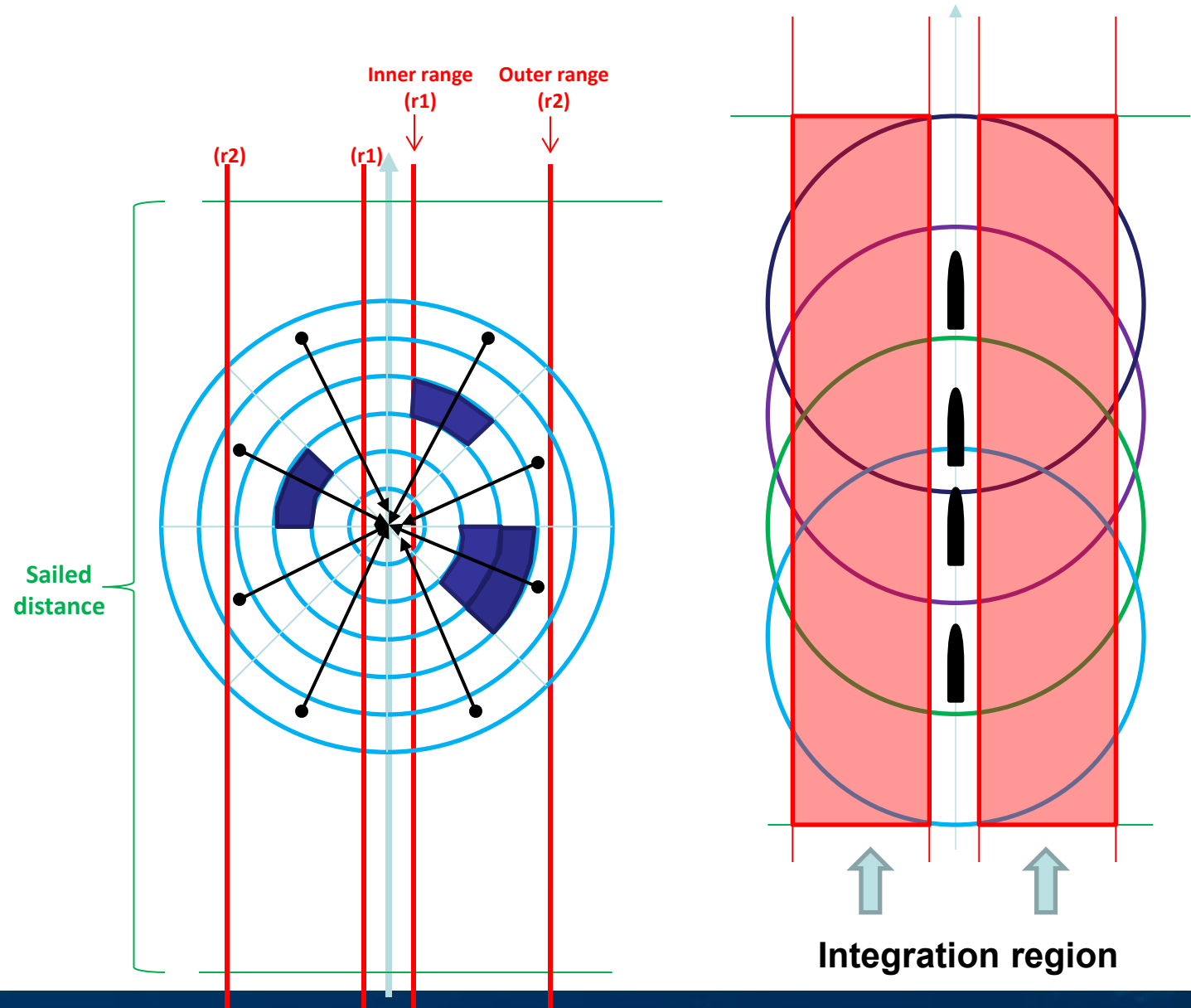
No depth resolution



One value for each ping

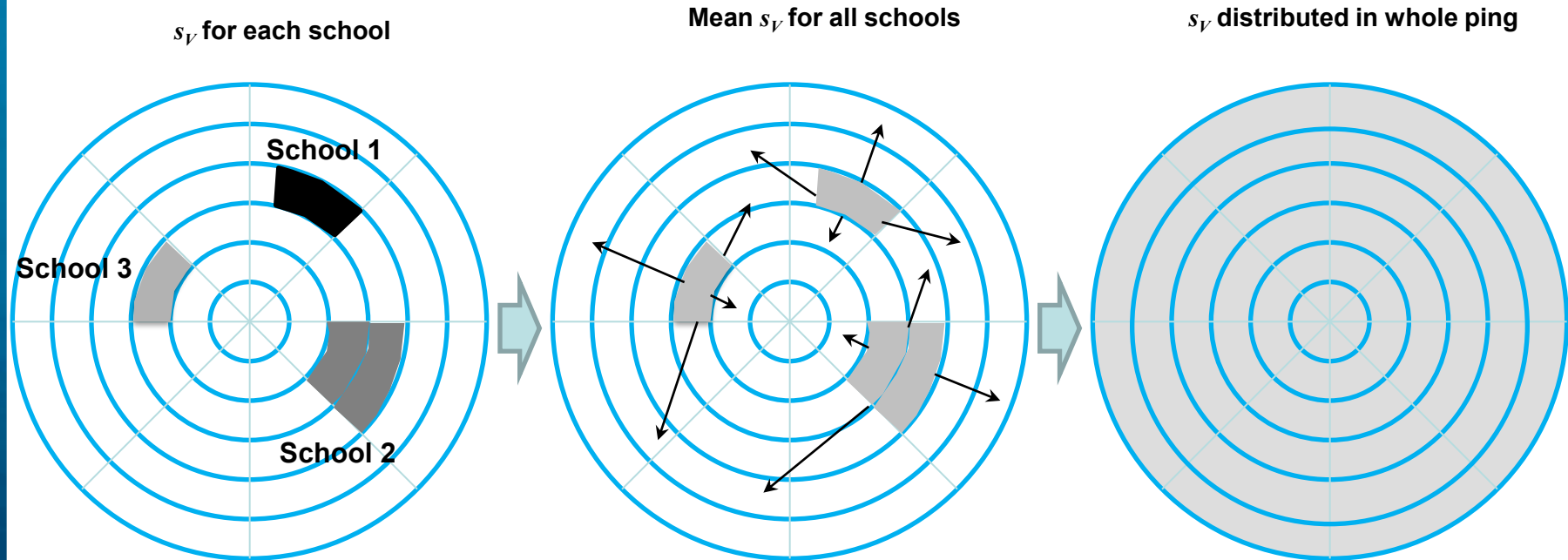
$$s_a = \int_{r1}^{r2} s_v dr$$

2. Echo integration of sonar data



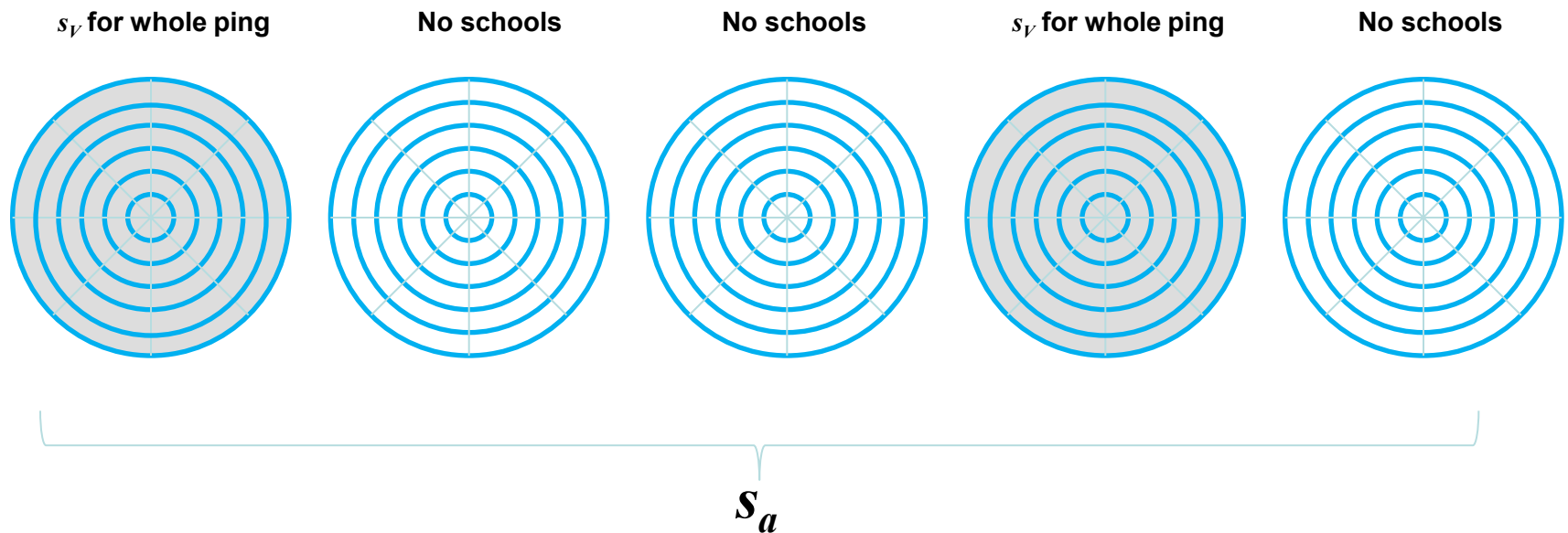
2. Echo integration of sonar data

Obtain a s_V value for each ping that contains one or more schools



2. Echo integration of sonar data

Obtain s_a for all pings (with or without schools) along the survey



2. Echo integration of sonar data

Sonar SX90

$$\bar{s}_{V_j} = \frac{\sum_{i=1}^n s_{V_i}}{n} \quad \text{mean school } s_v \text{ per ping}$$

$$c_j = \frac{\sum_{i=1}^n a_{s_i}}{a_b} \quad \text{Correction factor per ping}$$

$$s_{a_j} = \bar{s}_{V_j} c_j dz \quad s_a \text{ per ping with schools}$$

$$S_a = \frac{\sum_{j=1}^p s_{a_j}}{T} \quad s_a \text{ in all pings}$$

$$S_A = 4\pi 1852^2 S_a$$

$$N = \frac{S_A}{\sigma_{sp}} A \quad \text{total number of fish}$$

j = ping from 1 to p i = school in ping j

s_{V_i} = volume backscattering coefficient per school

n = total number of schools per ping

a_{s_i} = school area

a_b = all beams area

T = total number of pings

A = survey area

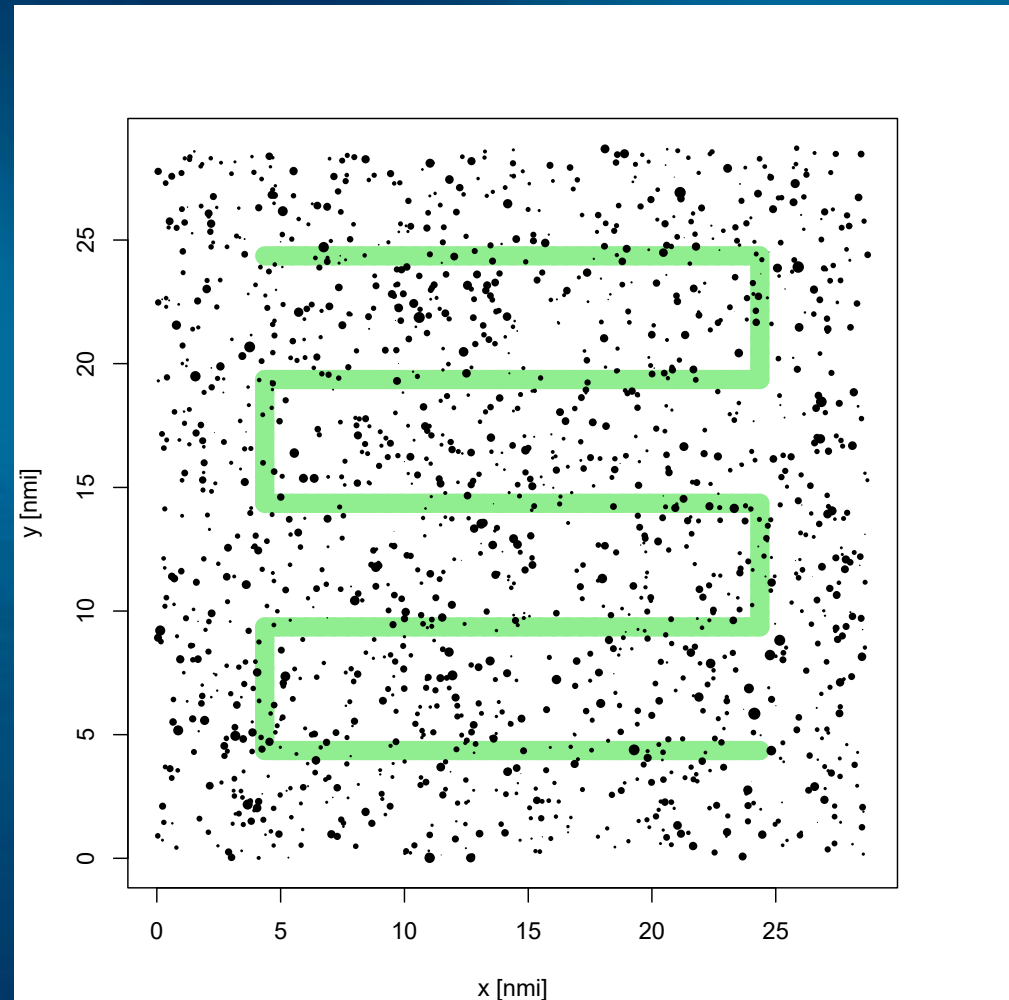
σ_{sp} = spherical scattering cross-section

dz = depth range

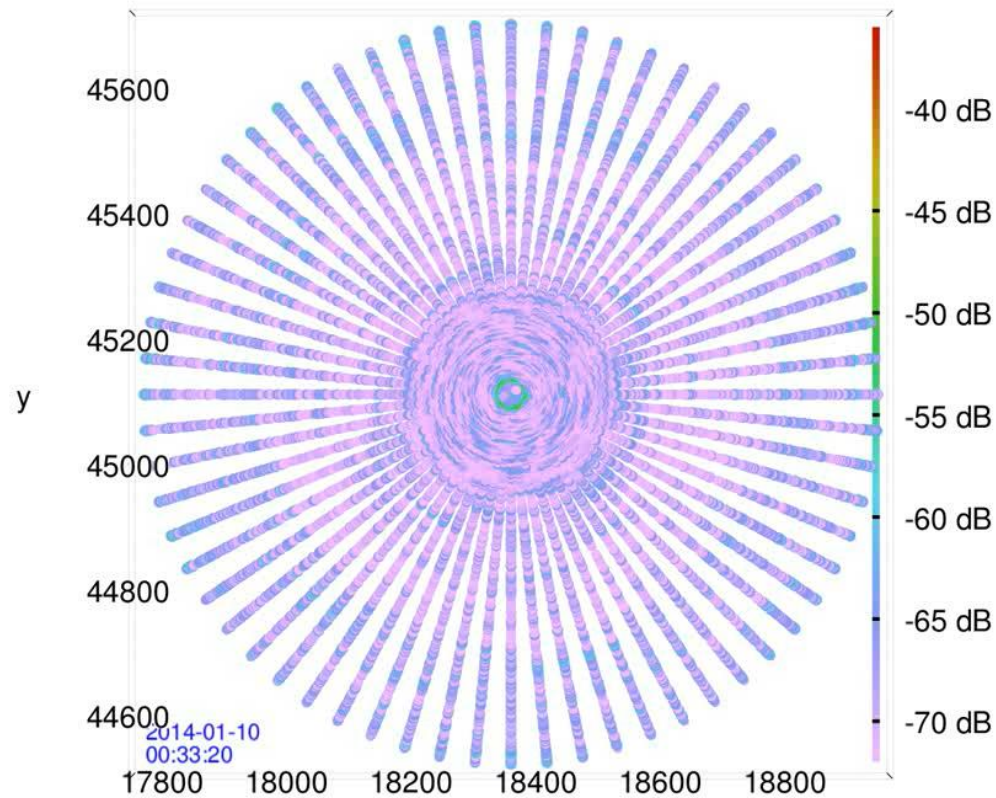
Simulation of SX90 sonar during surveying of herring schools

Simulation setup

- 20 × 20 nmi survey area (cruise track in green, schools in black). Total area with fish: $A = 824.6 \text{ nmi}^{-2}$.
- On average one school in the observation volume, average packing density 1 fish per m^3
- Gamma distributed school depths with mean 40 m and 5 % and 95 % quantile at 63 and 22 m
- Average school size: 50 × 50 × 10 m (ellipsoids)
- Omnidirectional fish



Simulation of SX90 sonar during surveying of herring schools



NASC values for sonar and echo sounder

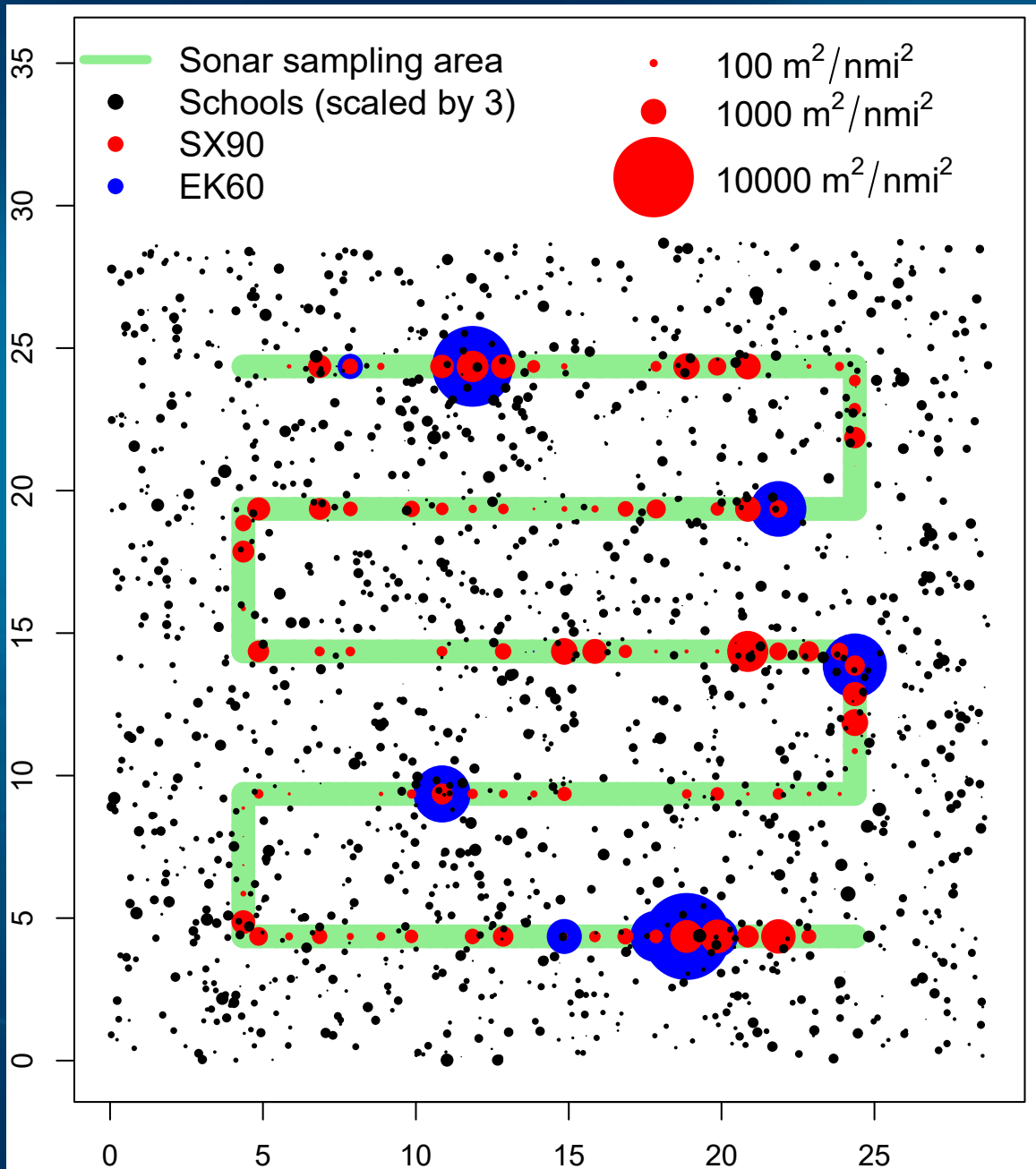
SX90 $\approx$ 70 red circles

EK60 $\approx$ 7 blue circles

$$N_{true} = 1.74E8$$

$$N_{SX90} = \frac{335.5}{2.56E-3} 824.6 = 1.08E8$$

$$N_{EK60} = \frac{416.6}{2.56E-3} 824.6 = 1.34E8$$



Issues

- Vertical distribution and detection probability. Should these be included in the echo integration?
- School area correction (not too important for echo integration)
- Target strength: Side aspect TS should be used instead of dorsal aspect
- Directionality: Average TS over all incidence angles
- Equivalent beam angle validity (Vatnehol and Handegard, J. Acoust. Soc. Am. 144 (4), October 2018)



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